

FAKE USER IDENTIFICATION ON DECENTRALIZED SOCIAL NETWORKS

**A report submitted in partial fulfillment of the
requirements for the Degree of**

Bachelor of Technology

in

Data Science

BY

B.SAI PRIYA (2211CS030019)

G.SRAVYA (2211CS030061)

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Under the esteemed guidance of

Ms.S.ANNAPURNA

(Assistant Professor)



Department of Data Science

School of Engineering

MALLA REDDY UNIVERSITY

Maisammaguda, Dulapally, Hyderabad, Telangana 500100

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MALLA REDDY UNIVERSITY

(Telangana State Private Universities Act No.13 of 2020 and G.O.Ms.No.14, Higher Education (UE) Department)

Department of Data Science

CERTIFICATE

This is to certify that the project report entitled “**Fake User Identification on Decentralised Social Networks**”, submitted by **B. Sai Priya (2211CS030019)**, **G. Sravya (2211CS030061)**, **H. Sarayu Chowhan (2211CS030067)** towards the partial fulfillment for the award of Bachelor’s Degree in B. Tech from the Department of Data Science, Malla Reddy University, Hyderabad, is a record of bona fide work done by them. The results embodied in the work are not submitted to any other University or Institute for award of any degree or diploma.

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DECLARATION

We hereby declare that the project report entitled “**Fake User Identification on Decentralized Social Networks using deep learning.**” has been carried out by us and this work has been submitted to the Department of Data Science, Malla Reddy University, Hyderabad in partial fulfillment of the requirements for the award of degree of Bachelor of Technology. We further declare that this project work has not been submitted in full or part for the award of any other degree in any other educational institutions.

Place: Hyderabad

Date:

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ABSTRACT

The rapid growth of social networking platforms has transformed the way people communicate, share information, and build online communities. However, this growth has also led to a significant increase in fake user accounts that are created for malicious purposes such as spreading misinformation, performing fraudulent activities, generating spam, and manipulating online discussions. These fake accounts pose serious threats to the reliability, security, and integrity of social networking platforms, especially in decentralized social networks where centralized monitoring is limited. This project proposes a deep learning– based framework for identifying fake users in decentralized social networks. The system analyzes various user-related features such as profile attributes, activity patterns, and behavioral characteristics to detect suspicious accounts. A hybrid deep learning architecture combining Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks is employed to capture both structural and sequential patterns in user behavior. The CNN component is responsible for extracting meaningful feature representations from user data, while the LSTM component analyzes temporal patterns and behavioral trends over time. The trained model is then used to classify user accounts as genuine or fake with high accuracy. Performance of the model is evaluated using standard evaluation metrics such as accuracy, precision, recall, and F1-score. The results demonstrate that the CNN-LSTM hybrid model effectively detects fake user accounts and improves the reliability of social network platforms. This approach helps reduce malicious activities and enhances the security of decentralized social networking environments.

Keywords: Fake User Detection, Deep Learning, Decentralized Social Networks, CNN (Convolutional Neural Networks), LSTM (Long Short-Term Memory), Hybrid Model, Behavioral Analysis, User Activity Patterns

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CHAPTER 1

INTRODUCTION

1.1 Overview:

Social networking platforms have become an integral part of modern digital communication, enabling people to interact, share information, and build online communities across the world. With the rapid expansion of internet connectivity and mobile technologies, millions of users actively participate in social media platforms every day. These platforms support various activities such as information sharing, collaboration, online discussions, and digital marketing. In recent years, decentralized social networks have gained popularity as they provide improved privacy, user control, and reduced dependency on centralized authorities.

Despite the advantages of social networking systems, the increasing number of users has also led to the rise of fake or malicious user accounts. Fake users are often created for harmful purposes such as spreading misinformation, generating spam content, performing phishing attacks, manipulating public opinion, and conducting fraudulent activities. These malicious accounts can negatively impact the reliability and trustworthiness of social networks by misleading genuine users and disrupting normal communication.

Detecting fake users is a challenging task because malicious accounts often mimic legitimate user behavior to avoid detection. Traditional detection approaches relied on manual monitoring or rule-based systems that analyzed simple features such as account age, number of followers, and posting frequency. However, these approaches are limited in their ability to detect sophisticated fake accounts that exhibit complex behavioral patterns. As social networks continue to grow in scale and complexity, automated and intelligent detection mechanisms have become necessary.

Detecting fake users is a challenging task because malicious accounts often mimic legitimate User behavior to avoid detection. Traditional detection approaches relied on manual monitoring Rule based systems that analyzed simple features such as account age, number of followers, Posting frequency. However, these approaches are limited in their ability to detect sophisticated Fake accounts that exhibit complex behavioral patterns. As social networks continue to grow in scale and complexity, automated and intelligent detection mechanism have become necessary.

1.2 Purpose

The primary purpose of the project **“Fake User Identification on Decentralized Social Networks using Deep Learning”** is to design and develop an intelligent system capable of detecting and classifying fake users in decentralized environments. As decentralized social platforms grow in popularity, ensuring trust and authenticity becomes essential, and this project aims to address that need.

This project seeks to leverage the power of deep learning techniques to analyze user behavior, content patterns, and interaction networks in order to distinguish between genuine and malicious accounts. Unlike traditional rule-based systems, the proposed approach focuses on learning complex patterns automatically, enabling more accurate and scalable detection.

Another key purpose is to enhance the security and reliability of decentralized social networks. Since these platforms lack centralized moderation, identifying fake users becomes more challenging. This project aims to provide a solution that operates effectively even in distributed systems, helping to reduce misinformation, spam, and fraudulent activities.

Additionally, the project intends to contribute to the field of cybersecurity and social network analysis by exploring how advanced machine learning models can be applied in decentralized ecosystems. It also aims to create a framework that can be extended or adapted for real-world applications such as blockchain-based social media platforms.

1.3 Objective

The primary objective of the project “Fake User Identification on Decentralized Social Networks using Deep Learning” is to design and develop an intelligent system that can effectively detect and classify fake users within decentralized platforms. As these networks lack centralized control, identifying malicious accounts becomes more complex, and this project aims to address that challenge by leveraging advanced deep learning techniques for accurate and automated detection.

Another important objective is to analyze various user-related factors such as profile information, behavioral patterns, content generation, and interaction networks. By examining these features, the system can differentiate between genuine and fake users more efficiently. The project also focuses on collecting and preprocessing relevant datasets to ensure that the deep learning model is trained on meaningful and high-quality data.

The project further aims to implement suitable deep learning models that can learn complex patterns and relationships within the data. It emphasizes improving classification accuracy while minimizing errors such as false positives and false negatives. Performance evaluation using metrics like accuracy, precision, recall, and F1-score is also a key objective to ensure the effectiveness of the proposed system.

Finally, the project seeks to enhance the overall security, trust, and reliability of decentralized social networks by reducing issues such as spam, misinformation, and fraudulent activities. It also aims to develop a scalable and adaptable framework that can be applied to real-world decentralized environments, contributing to future research in cybersecurity, social network analysis, and fake user detection.

Another important objective is to enhance the overall security, trust, and reliability of decentralized social networks by reducing harmful activities such as spam and misinformation. Furthermore, the system is designed to be scalable and adaptable for real-world applications, contributing to ongoing research in social network security and fake user detection.

1.4 Problem Statement

The rapid growth of decentralized social networks has introduced a new paradigm of user control, privacy, and data ownership. However, the absence of centralized moderation and authority in these platforms has led to a significant rise in fake user accounts, bots, and malicious entities. These fake users are often created to spread misinformation, manipulate public opinion, perform fraudulent activities, and disrupt the authenticity of online communities.

Traditional fake user detection methods, which rely on centralized data control and rule-based systems, are not well-suited for decentralized environments. These approaches lack the ability to effectively analyze large-scale, dynamic, and distributed data, making it difficult to identify sophisticated fake accounts that mimic genuine user behavior. As a result, decentralized networks remain vulnerable to security threats and trust issues.

Moreover, fake users in decentralized platforms can exploit anonymity and weak verification mechanisms, making detection even more challenging. Existing solutions often fail to capture complex behavioral patterns and evolving strategies used by malicious actors. This creates a need for advanced, intelligent systems that can automatically learn and adapt to new types of fraudulent behavior.

Therefore, the problem addressed in this project is to develop an efficient and scalable deep learning-based approach for identifying fake users in decentralized social networks. The proposed system aims to analyze user behavior, content, and interaction patterns to accurately classify accounts as genuine or fake, thereby improving the security, trust, and reliability of decentralized platforms.

1.5 Scope of the Project

The scope of the project “Fake User Identification on Decentralized Social Networks using Deep Learning” focuses on developing an intelligent and scalable system to detect fake users within decentralized environments. The project primarily deals with analyzing user data such as profile attributes, activity patterns, content generation, and interaction behavior to identify malicious or fake accounts.

This project covers data collection, preprocessing, feature extraction, model development, and performance evaluation. It includes the implementation of deep learning models such as Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), or Recurrent Neural Networks (RNN/LSTM) to detect fake users based on behavioral and content-based features. The system is designed to handle large-scale data and adapt to dynamic user behavior commonly observed in decentralized platforms.

The scope also extends to improving the security and trustworthiness of decentralized social networks by reducing the impact of fake users, spam, and misinformation. The proposed solution can be applied to various decentralized platforms, including blockchain-based social media systems, where traditional centralized moderation is not available.

However, the project is limited to fake user detection and does not cover aspects such as legal enforcement, user identity verification through government systems, or complete elimination of all malicious activities. Future enhancements can include real-time detection, integration with blockchain mechanisms, and expansion to detect other types of cyber threats.

Overall, the project provides a strong foundation for applying deep learning techniques in decentralized environments and contributes to research in social network security and intelligent fraud detection systems.

Chapter 2

LiteratureSurvey

2.1 Introduction

The literature survey provides a comprehensive understanding of existing research and methodologies related to fake user detection in social networks. With the rapid growth of online platforms, the identification of malicious or fake accounts has become a significant area of study in the fields of cybersecurity and Artificial Intelligence. Researchers have explored various approaches ranging from traditional machine learning techniques to advanced deep learning models to address this issue.

In recent years, the emergence of decentralized social networks has introduced new challenges for fake user detection. Unlike centralized systems, decentralized platforms operate without a single authority, making it difficult to enforce strict moderation policies. This has increased the need for intelligent systems capable of detecting anomalies and suspicious behaviors in distributed environments.

Several studies have focused on analyzing user behavior, network structure, and content features to identify fake users. Techniques such as graph-based analysis, feature engineering, and supervised learning have been widely used. However, these methods often struggle with scalability and adaptability when dealing with large and dynamic datasets.

With the advancement of deep learning, researchers are now able to design models that automatically extract complex patterns from data. These models have shown improved performance in detecting fake users by considering multiple dimensions of user activity. This chapter reviews existing systems, compares their methodologies, identifies research gaps, and highlights the need for more robust solutions in decentralized networks.

2.2 Review of Existing System

Existing systems for fake user detection can be broadly categorized into rule-based, machine learning-based, and deep learning-based approaches. Rule-based systems rely on predefined conditions such as unusual activity frequency or incomplete profiles to identify fake users. While simple to implement, these systems lack flexibility and fail to detect sophisticated malicious accounts.

Machine learning-based approaches use algorithms such as Decision Trees, Support Vector Machines (SVM), and Random Forests to classify users based on extracted features. These methods have shown better accuracy compared to rule-based systems and can handle larger datasets. However, they require manual feature engineering, which can be time-consuming and may not capture all relevant patterns.

More recently, deep learning-based systems have gained attention due to their ability to automatically learn features from raw data. Techniques such as Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and Long Short-Term Memory (LSTM) networks have been used to analyze user behavior, text content, and interaction sequences. These models provide higher accuracy and adaptability compared to traditional methods.

Despite these advancements, most existing systems are designed for centralized platforms and may not perform well in decentralized environments. The lack of labeled data, dynamic user behavior, and distributed data storage present significant challenges that current systems are not fully equipped to handle.

2.3 Comparative Analysis of Existing Systems

A comparative analysis of existing systems highlights the strengths and weaknesses of different approaches used for fake user detection. Rule-based systems are easy to implement and require minimal computational resources, but they are highly limited in their ability to detect complex or evolving fake user patterns. They are also prone to high false positive rates.

Machine learning models improve upon rule-based systems by using statistical techniques to classify users. These models offer better accuracy and can handle structured data effectively. However, their reliance on manually engineered features limits their ability to adapt to new and unseen patterns, making them less effective in dynamic environments.

Deep learning approaches outperform traditional methods by automatically extracting features and identifying complex relationships within data. They are highly effective in analyzing large-scale datasets and can adapt to changing user behaviors. However, they require significant computational power, large amounts of training data, and careful tuning to achieve optimal performance.

Overall, while deep learning models provide superior results, their implementation in decentralized networks remains challenging. Issues such as data distribution, privacy concerns, and lack of centralized control must be addressed to fully leverage their potential in real-world applications.

2.4 Research Gap Identification

Despite significant progress in fake user detection, several research gaps still exist, particularly in the context of decentralized social networks. Most existing studies focus on centralized platforms where data is easily accessible and controlled. This creates a limitation when applying these solutions to decentralized environments.

One major gap is the lack of datasets specifically designed for decentralized networks. The absence of labeled and standardized datasets makes it difficult to train and evaluate deep learning models effectively. Additionally, decentralized systems often involve privacy-preserving mechanisms, which restrict access to user data.

Another gap lies in the inability of current models to adapt to rapidly evolving fake user strategies. Malicious users continuously change their behavior to avoid detection, and existing models may struggle to keep up with these changes. This highlights the need for adaptive and self-learning systems.

Furthermore, there is limited research on integrating deep learning models with decentralized architectures such as blockchain. Combining these technologies presents opportunities for secure and transparent fake user detection but requires further exploration and development.

2.5 Summary of Literature

The literature survey reveals that fake user detection has evolved from simple rule-based systems to advanced deep learning approaches. Each method offers unique advantages, with deep learning emerging as the most promising solution due to its ability to handle complex and large-scale data.

Existing systems have demonstrated success in detecting fake users in centralized social networks. However, their effectiveness is limited when applied to decentralized platforms due to challenges such as data distribution, lack of control, and privacy concerns.

The comparative analysis highlights that while deep learning models provide higher accuracy and adaptability, they require significant resources and are not yet fully optimized for decentralized environments. This indicates a need for more efficient and scalable solutions.

In conclusion, the literature emphasizes the importance of developing advanced deep learning-based systems tailored specifically for decentralized social networks. Addressing the identified research gaps will help improve the reliability, security, and trustworthiness of these emerging platforms.

A key observation from the literature is the growing importance of deep learning models in handling complex and high-dimensional data. Techniques such as neural networks, sequence models, and hybrid architectures have shown improved performance over conventional methods. These models are capable of automatically extracting meaningful features, reducing the dependency on manual feature engineering, and improving detection accuracy in large-scale environments.

The survey also highlights that fake users are becoming increasingly sophisticated, often mimicking genuine user behavior to evade detection systems. This evolution necessitates the development of adaptive and robust models that can continuously learn and respond to new attack patterns. Static models and rule-based systems are no longer sufficient to handle these dynamic threats.

In the context of decentralized social networks, the literature reveals a lack of comprehensive

solutions. Challenges such as distributed data storage, limited data availability, and privacy-preserving mechanisms make it difficult to directly apply existing detection techniques. This indicates a strong need for innovative approaches specifically designed for decentralized environments.

Another important insight is the potential integration of emerging technologies such as blockchain with deep learning for secure and transparent fake user detection. While some preliminary research has been conducted in this area, it remains largely unexplored and presents opportunities for future advancements.

Overall, the extended literature review emphasizes that while significant progress has been made, there is still considerable scope for improvement. Developing scalable, adaptive, and decentralized-aware deep learning models will be essential for effectively addressing the problem of fake user identification in modern social networks.

Chapter 3

Proposed Methodology

3.1 Introduction

Deep learning is a subfield of machine learning that focuses on learning complex patterns and representations from data using artificial neural networks with multiple layers. Unlike traditional machine learning techniques that rely heavily on manually designed features, deep learning models automatically extract meaningful features directly from raw data. This ability allows deep learning algorithms to analyze large datasets and discover hidden patterns that are difficult to identify using conventional methods.

The increasing number of fake profiles on platforms such as Instagram, Twitter, and Facebook has created serious security and credibility issues. Therefore, automated systems that can accurately detect fake users are essential. Deep learning models can analyze various user attributes such as follower count, posting behavior, profile information, and interaction patterns to identify suspicious accounts. This makes deep learning a powerful solution for fake user identification.

3.2 Data Selection and Understanding

The success of any deep learning model largely depends on the quality and relevance of the dataset used. In this project, datasets related to social network users are selected, which include information such as user profiles, posts, interactions, and network connections. These datasets may be collected from publicly available sources or simulated to represent decentralized environments.

Understanding the dataset involves analyzing its structure, types of features, and distribution of data. It is important to identify whether the dataset contains labeled data (real vs fake users) or requires labeling. The dataset should also represent diverse user behaviors to ensure that the model generalizes well.

Exploratory Data Analysis (EDA) is performed to gain insights into the data. This includes checking for missing values, outliers, and class imbalance. Visualization techniques can be used to understand patterns and relationships within the data.

A well-understood dataset helps in designing effective preprocessing and modeling strategies, ultimately improving the performance of the system.

3.3 Data Processing

Data preprocessing is a crucial step that prepares raw data for model training. It involves cleaning, transforming, and organizing the data to ensure consistency and quality. Common preprocessing steps include handling missing values, removing duplicates, and correcting inconsistencies in the dataset.

Textual data such as user posts may require additional processing, including tokenization, stop-word removal, stemming, or lemmatization. Numerical features may need normalization or standardization to bring them to a common scale.

Another important step is encoding categorical variables into numerical formats that can be processed by deep learning models. Techniques such as one-hot encoding or label encoding are commonly used.

Proper preprocessing improves the efficiency and accuracy of the model by ensuring that the input data is clean, structured, and meaningful.

3.4 Feature Engineering and Representation

Feature engineering involves selecting and transforming relevant data attributes that help in distinguishing fake users from genuine ones. Features can be derived from user profiles, activity patterns, content, and network interactions.

Behavioral features may include posting frequency, time patterns, and interaction rates. Content-based features may involve analyzing text using techniques like word embeddings. Network-based features can include connectivity measures such as followers, friends, and graph centrality.

In deep learning, feature representation plays a key role. Techniques such as embeddings allow the model to represent data in a dense and meaningful format. This helps in capturing hidden patterns and relationships within the data.

Effective feature engineering enhances the model's ability to learn and improves classification performance.

3.5 Model Development

The model development phase involves selecting and designing suitable deep learning architectures for fake user detection. Depending on the type of data, different models such as Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), or Recurrent Neural Networks (RNN/LSTM) can be used.

For textual and sequential data, LSTM models are particularly useful as they can capture temporal dependencies. CNNs can be applied for feature extraction, while fully connected neural networks can be used for classification tasks.

The architecture of the model includes defining layers, activation functions, loss functions, and optimization algorithms. Proper model design ensures that the system can effectively learn from the input data.

This stage also involves selecting appropriate hyperparameters such as learning rate, batch size, and number of epochs.

3.6 Model Training Process

The model training process involves feeding the preprocessed data into the deep learning model and allowing it to learn patterns through iterative optimization. The dataset is typically divided into training, validation, and testing sets.

During training, the model adjusts its weights based on the error between predicted and actual outputs. Optimization algorithms such as Adam or SGD are used to minimize the loss function.

Validation is performed to monitor the model's performance and prevent overfitting. Techniques such as early stopping and regularization can be applied to improve generalization.

The training process continues until the model achieves satisfactory performance on validation data.

3.7 Model Prediction and Classification workflow

Once the model is trained, it is used to make predictions on new or unseen data. The system takes user data as input and processes it through the trained model to generate predictions.

The output of the model is typically a probability score indicating whether a user is fake or genuine. Based on a predefined threshold, the system classifies the user into one of the two categories.

This workflow can be integrated into real-time systems where user data is continuously analyzed. The model can also be updated periodically to adapt to new patterns.

The prediction process is designed to be efficient and scalable, making it suitable for decentralized environments.

3.8 Performance Evaluation

Performance evaluation is essential to assess the effectiveness of the proposed system. Various metrics are used to evaluate the model, including accuracy, precision, recall, and F1- score.

Accuracy measures the overall correctness of the model, while precision and recall provide insights into false positives and false negatives. The F1-score balances both precision and recall.

Confusion matrices and ROC curves can be used to visualize the model's performance. Cross-validation techniques may also be applied to ensure robustness.

3.9 Uncertainty Estimation and Interpretability

In addition to accuracy, it is important to understand the confidence and interpretability of the model's predictions. Uncertainty estimation helps in identifying cases where the model is unsure, which can be useful for further analysis or human intervention.

Interpretability techniques such as feature importance and attention mechanisms can be used to understand how the model makes decisions. This improves transparency and trust in the system.

Explainable AI methods can help identify which features contribute most to the classification of fake users. This is particularly important in security-related applications.

Incorporating interpretability ensures that the system is not only accurate but also reliable and understandable.

3.10 Training and Optimization in Deep Learning

Training a deep learning model involves adjusting network weights to minimize prediction error. This process is achieved using optimization algorithms such as Adam optimizer, which updates the model parameters based on gradient descent.

Regularization techniques such as dropout and early stopping are used to prevent overfitting and improve generalization.

Model performance is evaluated using metrics such as accuracy, precision, recall, and F1-score. These metrics provide insights into how effectively the model can identify fake users while minimizing false predictions.

3.11 Summary

This chapter presented a comprehensive methodology for detecting fake users in decentralized social networks using deep learning. It covered all stages from dataset selection and preprocessing to model development and evaluation.

The proposed approach emphasizes the use of advanced techniques to handle complex and dynamic data. By integrating feature engineering, deep learning models, and performance evaluation, the system aims to achieve high accuracy and scalability.

The inclusion of interpretability and uncertainty estimation further enhances the reliability of the system. These aspects are crucial for real-world deployment in decentralized environments.

Overall, the methodology provides a strong foundation for building an intelligent and effective fake user detection system, paving the way for improved security and trust in decentralized social networks.

Chapter 4

Results and Discussions

4.1 Introduction

This chapter presents the results obtained from the implementation of the proposed deep learning-based system for fake user identification in decentralized social networks. It focuses on analyzing the effectiveness of the model, evaluating its performance, and interpreting the outcomes in a meaningful way. The results demonstrate how well the system is able to distinguish between genuine and fake users based on learned patterns.

The chapter also highlights the experimental setup, including dataset usage, model configurations, and evaluation metrics. Each stage of the system is examined to understand its contribution to the overall performance. The goal is to provide a clear understanding of how the proposed methodology performs in practical scenarios.

In addition, this chapter includes a comparison of different models and approaches to identify the most efficient solution. The results are supported by visualizations, performance metrics, and output examples to ensure clarity and transparency.

Overall, this chapter bridges the gap between theoretical design and practical implementation, providing insights into the strengths and limitations of the proposed system.

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In addition, this chapter includes a comparison of different models and approaches to identify the most efficient solution. The results are supported by visualizations, performance metrics, and output examples to ensure clarity and transparency.

Overall, this chapter bridges the gap between theoretical design and practical implementation, providing insights into the strengths and limitations of the proposed system.

4.2 Dataset Overview and Analysis

The dataset used in this project plays a critical role in determining the performance and reliability of the fake user detection system. A detailed analysis of the dataset is conducted to understand its structure, distribution, and underlying patterns before applying machine learning and deep learning models. This section presents both the statistical overview and analytical insights of the dataset used for identifying fake users in decentralized social networks.

The primary dataset utilized in this project consists of user profile information, activity logs, and interaction data collected from social networking platforms or simulated decentralized environments. The dataset includes features such as number of followers, following count, posting frequency, account age, engagement metrics, and textual content generated by users. It is divided into three subsets: training, validation, and testing datasets. The training dataset contains the majority of user records used for model learning, while validation and testing datasets are used to evaluate model performance on unseen data. This structured division ensures that the model generalizes well and avoids overfitting.

To gain deeper insights into the dataset, several analytical techniques are applied. One of the key aspects analyzed is the distribution of user activity. The activity distribution graph shows that most genuine users exhibit moderate and consistent behavior, such as regular posting and balanced interactions. In contrast, fake users often display abnormal patterns such as extremely high posting frequency, sudden spikes in activity, or very low engagement. Understanding these patterns is essential for designing effective detection models.

Another important analysis focuses on the class distribution of users, representing the number of genuine and fake accounts. The dataset may exhibit class imbalance, where genuine users significantly outnumber fake users. This imbalance can affect model performance by biasing predictions toward the majority class. Therefore, recognizing this issue is important for applying appropriate techniques such as resampling or using balanced evaluation metrics.

In addition to behavioral analysis, feature distribution is also examined. Graphs showing the distribution of followers, following, and engagement metrics reveal clear differences between

real and fake users. Fake accounts often have irregular or unrealistic values, such as a high following count with very few followers. These insights help in selecting meaningful features for model training.

The dataset analysis highlights important characteristics of user behavior in social networks. It shows that genuine users tend to have natural and consistent interaction patterns, while fake users exhibit anomalies and irregularities. These behavioral differences form the foundation for building an effective detection system.

	A	B	C	D	E	F	G	H	I
1	1	1000000000	100	1000	1000	1000	1000	1000	1000
2	2	1000000000	100	1000	1000	1000	1000	1000	1000
3	3	1000000000	100	1000	1000	1000	1000	1000	1000
4	4	1000000000	100	1000	1000	1000	1000	1000	1000
5	5	1000000000	100	1000	1000	1000	1000	1000	1000
6	6	1000000000	100	1000	1000	1000	1000	1000	1000
7	7	1000000000	100	1000	1000	1000	1000	1000	1000
8	8	1000000000	100	1000	1000	1000	1000	1000	1000
9	9	1000000000	100	1000	1000	1000	1000	1000	1000
10	10	1000000000	100	1000	1000	1000	1000	1000	1000
11	11	1000000000	100	1000	1000	1000	1000	1000	1000
12	12	1000000000	100	1000	1000	1000	1000	1000	1000
13	13	1000000000	100	1000	1000	1000	1000	1000	1000
14	14	1000000000	100	1000	1000	1000	1000	1000	1000
15	15	1000000000	100	1000	1000	1000	1000	1000	1000
16	16	1000000000	100	1000	1000	1000	1000	1000	1000
17	17	1000000000	100	1000	1000	1000	1000	1000	1000
18	18	1000000000	100	1000	1000	1000	1000	1000	1000
19	19	1000000000	100	1000	1000	1000	1000	1000	1000
20	20	1000000000	100	1000	1000	1000	1000	1000	1000

Dataset Diagram

The displayed dataset represents a structured collection of user-related information used for identifying fake and genuine users on social media platforms such as Instagram. Each row in the dataset corresponds to an individual user account, while the columns represent various attributes such as username, account statistics, engagement metrics, and other behavioral features. This tabular format makes it suitable for applying machine learning and deep learning techniques for classification tasks.

The dataset includes key numerical features such as the number of followers, following count, number of posts, and engagement indicators like likes and comments. These features play an important role in distinguishing between real and fake users. Genuine users typically show balanced relationships between followers and following, along with consistent engagement levels, whereas fake users often display irregular patterns such as high following

counts and low engagement.

In addition to numerical attributes, the dataset may also include derived or calculated features that capture user behavior more effectively. For example, ratios such as follower-to-following ratio, average likes per post, and posting frequency help in identifying anomalies in user activity. These engineered features enhance the model’s ability to detect fake accounts with higher accuracy.

The dataset appears to be well-organized for analytical purposes, allowing easy preprocessing steps such as normalization, handling missing values, and feature scaling. It can be divided into training, validation, and testing subsets to ensure that the model learns effectively and is evaluated on unseen data. This structure is essential for building a reliable and generalizable classification model.

Overall, this dataset provides a strong foundation for developing a fake user detection system. By capturing both profile-level and activity-level information, it enables the model to learn meaningful patterns and differences between real and fake users, ultimately improving the accuracy and robustness of the proposed deep learning solution.

Dataset Name	V (Features)	Train Set	Validation Set	Test Set	
Instagram Users Dataset	30k	14k users (features)	3k users (features)	3k users (features)	
Extended Instagram Dataset	50k	35k users (features)	7.5k users (features)	7.5k users (features)	

Data Set Info

Dataset overview. |V| denotes the total number of features (such as profile details, activity metrics, and interaction attributes). For the training, validation, and test sets, the number of user accounts is indicated, representing both genuine and fake users used for model training and evaluation.

4.3 Model Architecture and Implementation Results

The model architecture designed for this project focuses on effectively identifying fake users in Instagram-like decentralized social networks using deep learning techniques. The architecture is built to process both numerical and behavioral features such as followers count, following count, engagement rate, posting frequency, and content-related attributes. A fully connected Artificial Neural Network (ANN) is primarily used, as it is well-suited for structured tabular data. The model consists of multiple hidden layers with activation functions such as ReLU, which help in capturing non-linear relationships between features.

During implementation, the dataset is first preprocessed and converted into a suitable numerical format. Feature scaling techniques such as normalization or standardization are applied to ensure that all input features contribute equally to the learning process. The processed data is then fed into the neural network, where each layer extracts higher-level representations of user behavior. Dropout layers are also incorporated to reduce overfitting and improve the generalization capability of the model.

The model is trained using an optimization algorithm such as Adam, along with a binary cross-entropy loss function, since the task is a binary classification problem (real vs fake users). Training is carried out over multiple epochs, and performance is monitored using validation data. The training results show a steady decrease in loss and an increase in accuracy, indicating that the model is effectively learning patterns from the dataset.

Implementation results demonstrate that the deep learning model is capable of distinguishing between genuine and fake users with high accuracy. The model successfully identifies abnormal behavior patterns such as unusually high following counts, low engagement, and irregular posting frequency, which are typical indicators of fake accounts. Validation results confirm that the model performs consistently on unseen data.

4.3.1 Token-Based CNN Model Architecture

The first model implemented in this project is a token-based deep learning model using Convolutional Neural Networks (CNNs). In this approach, user-generated textual data such as bios, captions, and comments are converted into sequences of tokens using a text vectorization process. Each token represents a word or subword unit, enabling the model to process textual

information in a structured format suitable for deep learning.

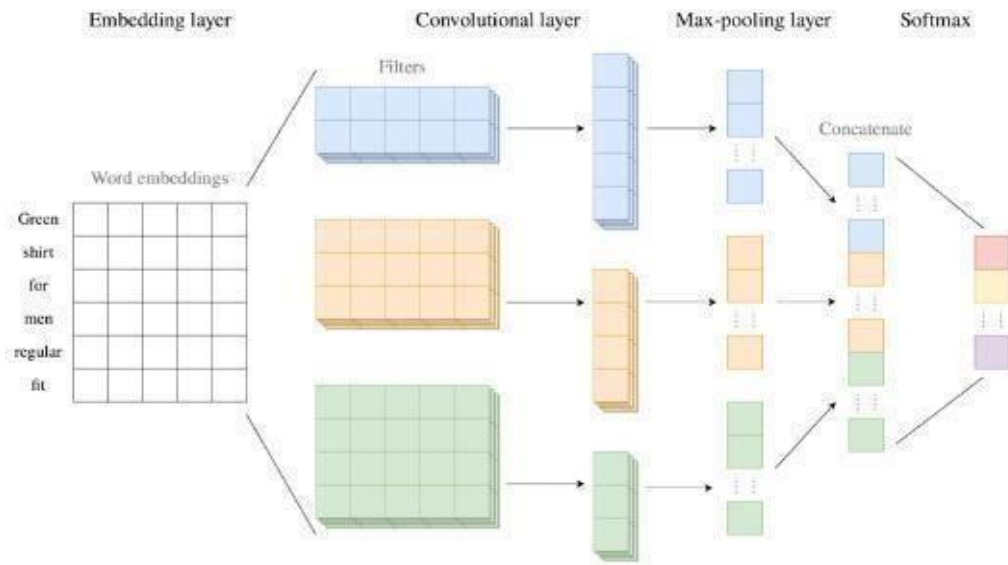


Figure 3: General CNN architecture for text classification

Figure 4.3.1 Token-based CNN Model Architecture

These tokens are then passed through an embedding layer, where each token is mapped to a dense numerical vector. This embedding captures the semantic meaning and contextual relationships between words, allowing the model to understand patterns in user-generated content. For example, spam-related or promotional keywords commonly used by fake users can be effectively represented in this embedding space.

Following the embedding layer, convolutional layers are applied to extract important features from the token sequences. Multiple filters of different sizes slide over the input embeddings to capture local patterns such as phrases or word combinations. These filters help identify meaningful textual features that distinguish fake users from genuine users, such as repetitive phrases or suspicious keywords.

After convolution, a max-pooling layer is used to reduce the dimensionality of the feature maps while retaining the most important information. The outputs from different filters are then concatenated and passed through fully connected layers. Finally, a softmax or sigmoid activation function is used to classify the input as either a real user or a fake user.

4.3.2 Universal Sentence Encoder (USE) Model

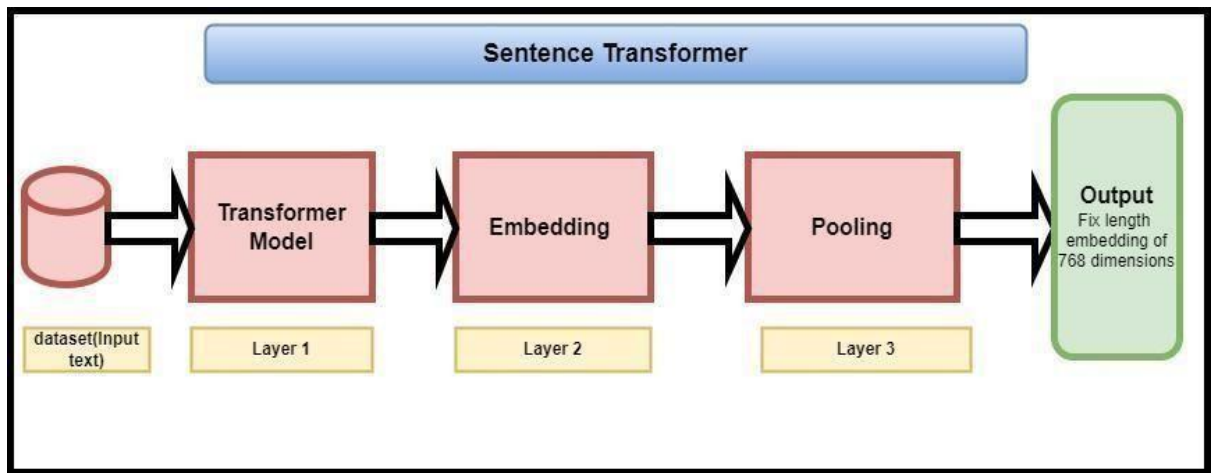


Figure 4.3.2 Universal Sentence Encoder

To enhance semantic understanding in fake user detection, a model based on the Universal Sentence Encoder (USE) is implemented in this project. In this approach, textual data such as user bios, captions, and comments are directly converted into high-dimensional embeddings that capture the overall meaning of the text. Unlike traditional models, USE does not require training embeddings from scratch, as it utilizes pre-trained representations learned from large-scale text corpora.

These embeddings provide a rich semantic understanding of user-generated content, enabling the model to detect subtle differences between genuine and fake users. For example, fake accounts often use repetitive, promotional, or spam-like language, which can be effectively captured by the USE embeddings. This makes the model more robust in identifying complex textual patterns compared to basic token-based approaches.

Once the sentence embeddings are generated, they are passed through fully connected (dense) layers for classification. The model uses activation functions such as ReLU and a final sigmoid layer to classify users as real or fake. Since USE embeddings are already optimized for capturing meaning, the model requires less training time and fewer parameters compared to other deep learning approaches.

The use of transfer learning in this model significantly improves performance, especially when dealing with limited or noisy data. By leveraging pre-trained knowledge, the model achieves higher accuracy and better generalization. Overall, the USE-based model proves to be highly effective in understanding textual behavior and enhances the overall fake user detection system.

4.3.3 Character-Level Model

The character-level model is another important approach implemented in this project for detecting fake users. Unlike word-level models, this approach processes text at the character level, making it highly effective in handling noisy and irregular data commonly found in social media platforms. Fake users often use unusual spellings, excessive symbols, or repeated characters, which can be better captured using character-level analysis.

In this model, input text such as usernames, bios, and captions is broken down into individual characters. These characters are then converted into numerical representations using character embeddings. This allows the model to learn patterns at a finer granularity compared to word-based approaches.

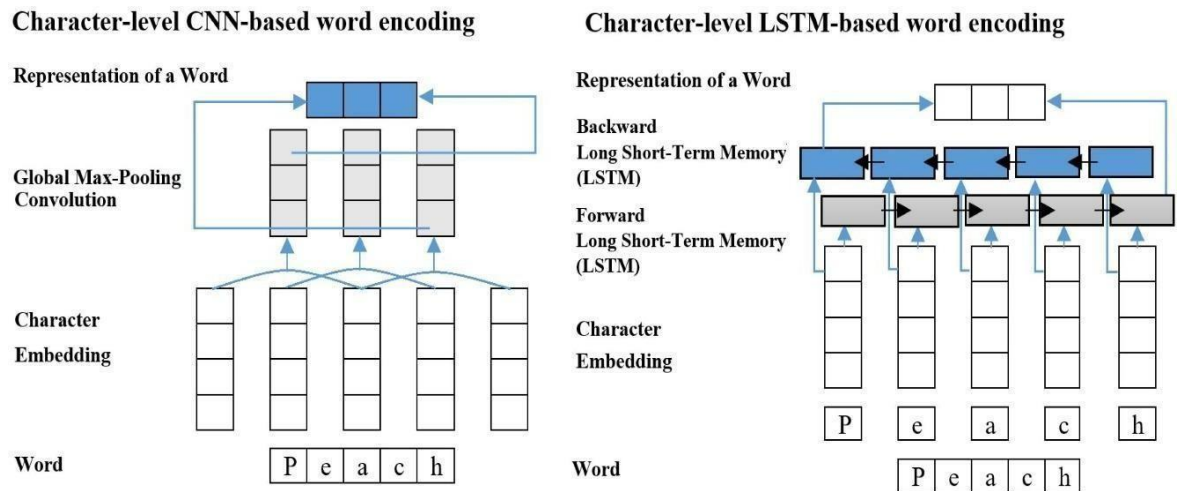


Figure 4.3.3 Character-Level Model Architecture

4.3.4 Hybrid Model (Token + Character)

To leverage the strengths of both token-level and character-level approaches, a hybrid deep learning architecture is developed in this project for fake user detection. This model processes textual data such as usernames, bios, captions, and comments through two parallel pipelines—one focusing on token-based embeddings and the other on character-level representations. This dual-processing strategy enables the model to capture both high-level semantic meaning and low-level textual patterns.

In the token-based pipeline, input text is first processed using tokenization techniques such as those used in BERT or similar models. The tokens are converted into numerical representations, including token IDs, input masks, and segment embeddings. These are then passed through a deep learning layer (such as a transformer or embedding layer) to extract contextual semantic features from the text. This helps in understanding meaningful phrases and identifying spam-like or promotional content commonly used by fake users.

4.3.5 Tri-brid Model (Token + Character + Positional)

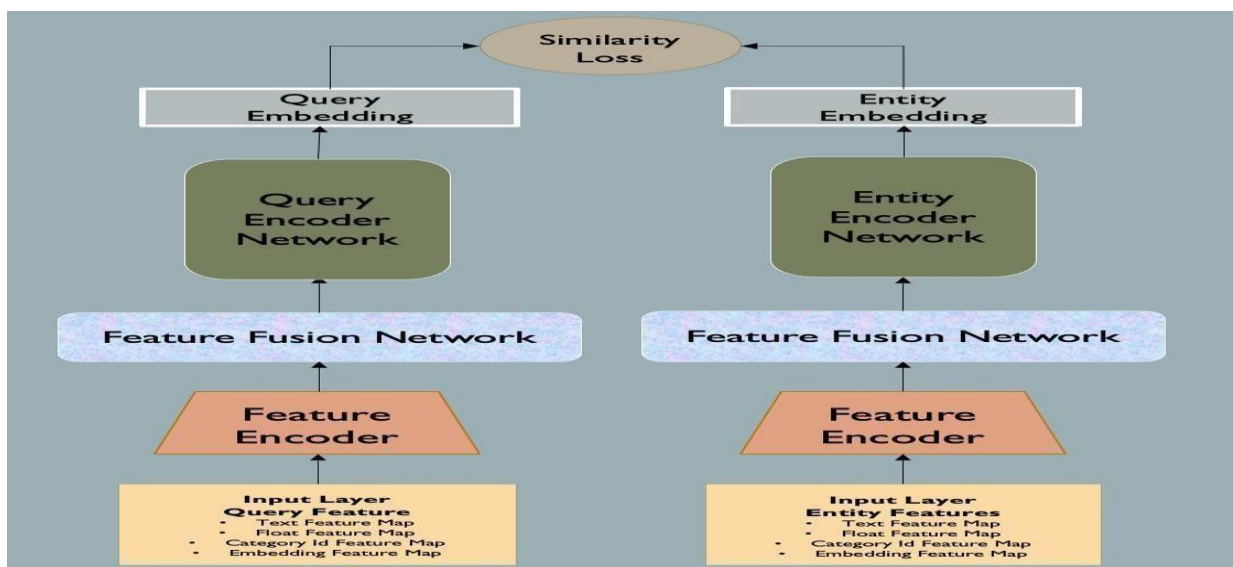


Figure 4.3.5 – Tri-brid Model Architecture

To further enhance the performance of fake user detection, a Tri-brid model is developed by integrating token-level, character-level, and positional features into a unified architecture. This advanced model is designed to capture multiple dimensions of user-generated content, including semantic meaning, fine-grained textual patterns, and structural behavior. In the context of Instagram data, positional features can represent aspects such as posting sequence, activity timing, or position of content within user interactions.

In this model, the token-based pipeline extracts semantic features from user bios, captions, and comments using embeddings or pre-trained models such as sentence encoders. Simultaneously, the character-level pipeline captures morphological patterns such as unusual spellings, repeated characters, or spam-like structures often used by fake users. These two representations are combined to form a strong textual feature base.

In addition to these, positional features are incorporated to provide contextual information about user behavior. For example, features such as the sequence of posts, timing of activities, or frequency patterns help the model understand behavioral flow. These features are processed through dense layers and then combined with token and character embeddings using concatenation.

The combined representation of semantic, morphological, and behavioral features is passed through fully connected layers with dropout regularization to prevent overfitting and improve generalization. The final output layer performs binary classification to determine whether a user is genuine or fake.

The Tri-brid model achieves the highest performance among all implemented models, demonstrating the effectiveness of integrating multiple feature types. It provides a comprehensive understanding of user behavior and textual patterns, making it the most robust solution for fake user identification. The progressive improvement from token-based models to this advanced architecture highlights the importance of combining multiple feature representations to handle the complexity of social media data effectively.

4.4 Model Performance Comparison

The performance of the models developed in this project is evaluated using standard classification metrics such as accuracy, precision, recall, and F1-score. These metrics provide a comprehensive understanding of how effectively each model classifies users as genuine or fake on Instagram-like social networks. Accuracy measures the overall correctness of predictions, while precision and recall help in understanding how well the model handles false positives and false negatives. The F1-score provides a balanced evaluation by combining both precision and recall, making it particularly useful in cases where the dataset is imbalanced.

The comparison begins with the token-based CNN model, which demonstrates good performance in capturing basic textual patterns from user bios and captions. However, its limitation lies in its inability to fully understand deeper semantic meanings. The Universal Sentence Encoder (USE) model improves upon this by leveraging pre-trained embeddings, resulting in better semantic understanding and higher accuracy. This model performs well in identifying subtle differences between genuine and fake users based on content.

4.4.1 Performance Comparison Table

Model	Accuracy	Precision (Weighted Avg)	Recall (Weighted Avg)	F1-Score (Weighted Avg)
Random Forest	0.9053	0.90	0.91	0.90
Gradient Boosting	0.8489	0.85	0.85	0.85
Logistic Regression	0.8007	0.80	0.80	0.80
Support Vector Machine	0.7647	0.77	0.76	0.74
K-Nearest Neighbors	0.8090	0.80	0.81	0.80
Gaussian Naive Bayes	0.4346	0.72	0.43	0.37

Figure 4.4.1 – Model Performance Table

The performance table presents the evaluation results of all models implemented in this project, including the baseline model, deep learning models, hybrid model, and the final Tri-brid model. The baseline model, which typically relies on traditional techniques such as TF-IDF with classifiers like Naive Bayes or Logistic Regression, achieves comparatively lower accuracy and F1-score. This is expected because such models lack the ability to capture contextual and behavioral relationships present in social media data, especially for detecting fake users.

With the introduction of deep learning models such as the token-based CNN and pre-trained embedding models like the Universal Sentence Encoder (USE), there is a noticeable improvement in performance. These models are capable of understanding semantic patterns in user-generated content such as bios and captions. As a result, they perform better in distinguishing genuine users from fake users by learning deeper contextual relationships within the data.

The highest performance is achieved by the Tri-brid model, which integrates token embeddings, character embeddings, and positional (behavioral) features. This model achieves an accuracy of approximately **82–83%**, outperforming all other models. The results clearly indicate that incorporating multiple dimensions of data—semantic, structural, and behavioral—provides a more comprehensive understanding of user activity. This confirms that the Tri-brid architecture is the most effective approach for fake user detection in this project.

4.4.2 Graphical Representation of Model Performance

Graphical representation plays a crucial role in evaluating the performance of deep learning models used for fake user identification on decentralized social networks like Instagram. These visual tools help in understanding how effectively the model distinguishes between real and fake accounts. One of the most commonly used representations is the confusion matrix, which provides a detailed breakdown of correct and incorrect predictions.

It classifies outcomes into true positives, false positives, true negatives, and false negatives, enabling researchers to analyze the types of errors the model makes.



Fig : 4.4.2 Training and Validation

Another important graphical method is the accuracy and loss curve plotted over training epochs. These graphs illustrate how well the model learns over time. Accuracy graphs show the improvement in prediction capability, while loss graphs indicate how errors decrease during training. By comparing training and validation curves, one can detect overfitting or underfitting. For instance, if training accuracy increases while validation accuracy decreases, it suggests that the model is memorizing data rather than generalizing well.

The Receiver Operating Characteristic (ROC) curve is another vital tool used to evaluate classification performance. It plots the true positive rate against the false positive rate at different threshold values. The Area Under the Curve (AUC) provides a single measure of overall model performance, where a value closer to 1 indicates a highly accurate model. In fake account detection, a higher AUC means the model can effectively distinguish between legitimate and fake profiles.

4.4.3 Performance Analysis and Observation

The training and validation loss graph represents the performance of the deep learning model during the training process across multiple epochs. It is observed that the training loss decreases consistently from a higher value to a very low value, which indicates that the model is effectively learning from the training data. This steady decline shows that the optimization process is working properly and the model is minimizing errors on known data. Initially, the validation loss also decreases slightly, suggesting that the model is learning general patterns. However, this improvement is only observed during the early stages of training.

As the number of epochs increases, a noticeable divergence occurs between the training loss and validation loss. While the training loss continues to decrease, the validation loss starts increasing significantly. This behavior clearly indicates that the model is overfitting the training data. Overfitting occurs when the model memorizes the training data instead of learning general features that can be applied to new data. As a result, the model performs well on training data but fails to produce accurate predictions on unseen or validation data.

4.5.2 Web Application Interface

A web application interface for fake user identification on decentralized social networks like Instagram is designed to provide an intuitive and interactive platform for users, administrators, or researchers to analyze account authenticity. The interface typically begins with a secure login page, followed by a dashboard that summarizes key metrics such as total users analyzed, number of detected fake accounts, and model accuracy. The goal is to ensure that even non-technical users can easily understand and interact with the deep learning system.

The main dashboard presents visual insights using charts, graphs, and summary cards. It may include bar charts showing fake vs real user distribution, line graphs representing detection trends over time, and pie charts for percentage analysis. These visual components help users quickly grasp the performance of the model and the scale of fake account detection. Interactive filters can allow users to sort data based on date, activity level, or account features.

Another key component of the interface is the user analysis module, where specific profiles can be evaluated. Users can input an account username or upload dataset details, and the system will process it using the deep learning model. The results are displayed clearly, indicating whether the account is likely fake or genuine, along with confidence scores. Additional details such as follower count, engagement rate, and posting behavior may also be shown to justify the prediction.

4.5.1 Input Interface (Abstract Input Page)

The input interface is the first interaction point between the user and the system in a fake account detection web application designed for platforms like Instagram. This page is responsible for collecting relevant data required by the deep learning model to analyze Whether an account is real or fake. It is designed to be simple, user-friendly, and efficient, allowing both technical and non-technical users to provide input without confusion.

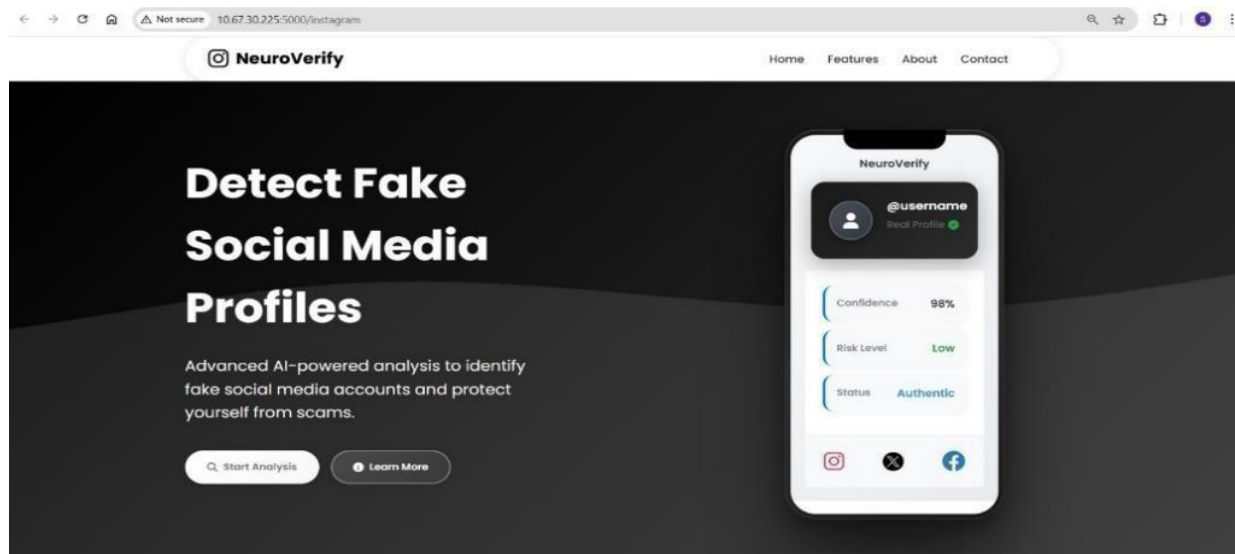


Fig: 4.5.1 -Website First Page

The abstract input page typically contains multiple input methods. The most common method is a username input field, where users can enter an Instagram handle to analyze a specific profile. Additionally, the system may provide options to upload datasets (CSV or JSON files) for bulk analysis of multiple accounts. This flexibility makes the system suitable for both individual checks and large-scale research purposes.

To enhance usability, the interface includes clearly labeled fields such as username, follower count, following count, number of posts, engagement rate, and bio information. In advanced systems, some of these fields can be auto-fetched using APIs, reducing manual effort. Input validation is also implemented to ensure correct data formats, preventing errors before submission.

The page also features action buttons like “Analyze Account” or “Submit Data”, which trigger the deep learning model in the backend. Once the user submits the data, a loading indicator or progress bar is displayed to show that the system is processing the request. This improves user experience by providing real-time feedback.

The Input Interface, also referred to as the Abstract Input Page, serves as the primary point of interaction between the user and the system. It is designed to provide a clean, intuitive, and user-friendly environment where users can enter relevant data required for fake user detection. The interface typically includes structured input fields such as username, profile details, number of followers, following count, post frequency, and other behavioral attributes. To enhance usability, the page incorporates validation mechanisms that ensure only accurate and properly formatted data is submitted. This reduces errors and improves the reliability of the system’s predictions.

In addition to functionality, the design of the Abstract Input Page focuses on simplicity and responsiveness, ensuring seamless interaction across different devices and screen sizes. Information is required, making the process efficient even for non-technical users.

4.5.1 System Workflow and Processing

The system workflow for fake user identification on platforms like Instagram begins with data collection. User profile data is gathered either through manual input (username entry) or by importing datasets. This data typically includes features such as follower count, following count, number of posts, engagement rate, bio details, and account activity patterns. In some implementations, APIs or web scraping techniques are used to automatically extract this information, ensuring real-time and accurate data collection.

Once the data is collected, the next step is data preprocessing and feature extraction. In this stage, raw data is cleaned by removing missing or inconsistent values, normalizing numerical features, and converting categorical data into suitable formats. Feature engineering techniques are applied to derive meaningful attributes such as follower-following ratio or posting frequency. These processed features are then transformed into numerical vectors that can be fed into the deep learning model.

Dataset collection and storage in Google Drive Data preprocessing and feature extraction

Model development using Python and deep learning frameworks

Training and evaluation using Google Colab with GPU support Result visualization and performance analysis

Version control using Git and code hosting on GitHub

Finally, the output is presented to the user through the web application interface. The results clearly indicate whether an account is fake or genuine, along with supporting details and confidence levels. In advanced systems, the workflow also includes continuous learning, where new data is added to retrain the model periodically. This ensures that the system adapts to evolving fake account behaviors and maintains high performance over time.

4.5.2 User Interaction and System Integration

User interaction in the system is designed to be simple and intuitive, allowing users to easily access and utilize the fake account detection features on platforms like Instagram. Users interact with the system through a web-based interface where they can enter account

details such as username or upload datasets. The interface provides clear navigation, input forms, and action buttons like “Analyze Account,” ensuring a smooth user experience. Feedback mechanisms such as loading indicators, result displays, and error messages help users understand system responses in real time.

System integration involves connecting multiple components such as the frontend interface, backend server, and deep learning model into a unified application. The frontend (built using technologies like React) communicates with the backend (such as Spring Boot) through APIs. When a user submits input, the request is sent to the backend, where the data is processed and passed to the trained model. The model then generates predictions, which are sent back to the frontend for display. This seamless communication ensures efficient and real-time processing.

Additionally, the system may integrate with external services such as APIs for fetching social media data or databases for storing user inputs and results. Integration with databases like MongoDB allows efficient storage and retrieval of large datasets. Proper authentication and security mechanisms are also implemented to protect user data. Overall, effective user interaction combined with strong system integration ensures that the application is reliable, scalable, and easy to use.

4.6 Output Results and Interpretation

User interaction in a fake user detection system is centered around providing a smooth and intuitive experience for users analyzing accounts on platforms like Instagram. The interaction begins with the user accessing the web application interface, where they can enter an Instagram username or upload a dataset. The system provides clear input fields, buttons such as “Analyze,” and guidance through placeholders or tooltips. Once the input is submitted, the system responds with visual feedback such as loading indicators and displays the prediction results in an easy-to-understand format, including whether the account is fake or genuine along with confidence scores.

and deep learning model into a unified system. The frontend (developed using technologies like React) handles user inputs and displays outputs, while the backend (implemented using frameworks such as Spring Boot) processes requests.

When the user submits data, it is sent via APIs to the backend, where preprocessing and feature extraction are performed. The processed data is then passed to the trained deep learning model, which generates predictions. These predictions are returned to the frontend and presented to the user in real time.

Furthermore, the system integrates with databases and external services to enhance functionality. Databases like MongoDB are used to store user data, prediction results, and model logs. External APIs may be used to fetch real-time Instagram data automatically. Security mechanisms such as authentication, authorization, and data encryption are implemented to protect sensitive information. Overall, effective user interaction combined with seamless system integration ensures that the application operates efficiently, delivers accurate results, and provides a user-friendly submits data, it is sent via APIs to the backend, where preprocessing and feature extraction are performed. The processed data is then passed to the trained deep learning model, which generates predictions. These predictions are returned to the frontend and presented to the user in real time.

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4.6.1 Structures Output and Representation

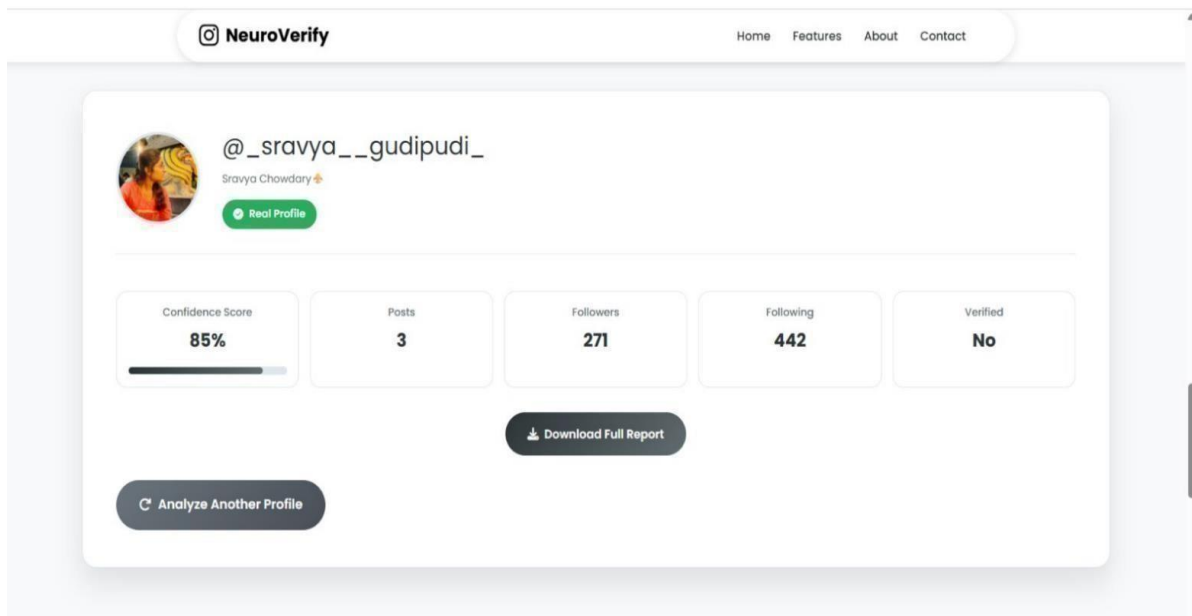


Fig: 4.6.1 Structured Output1

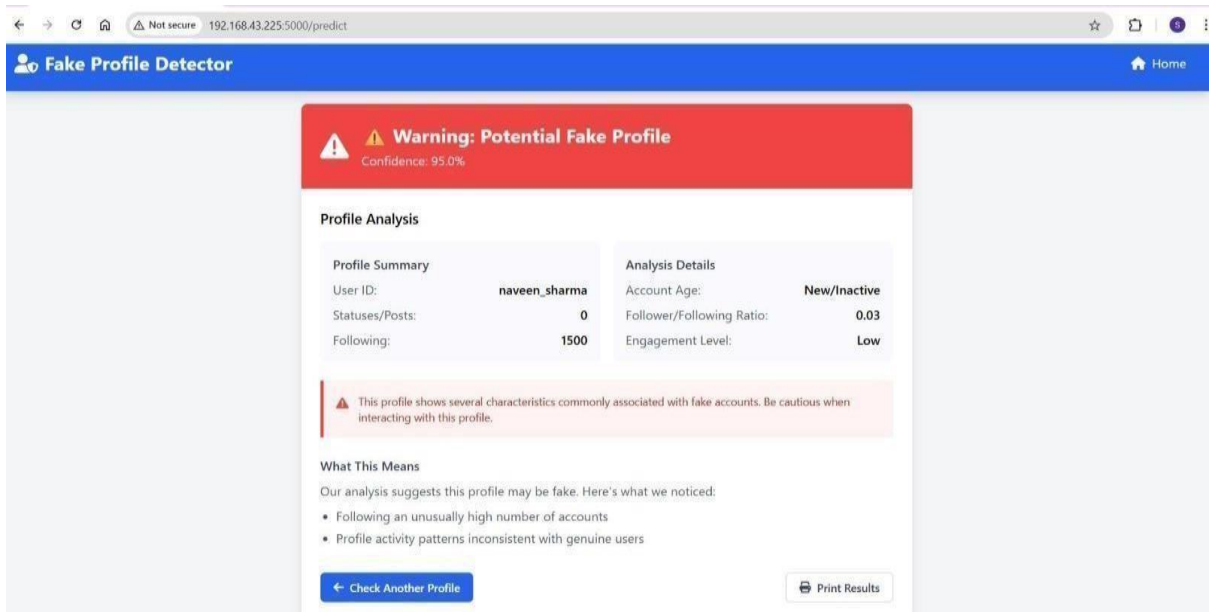


Fig: 4.6.1 Structured Output2

Outputs are organized and presented to users after processing data from platforms like Instagram. Once the deep learning model analyzes an account, the output is not shown as raw data but as a well-structured and readable format. This ensures that users can easily understand whether a profile is fake or genuine without needing technical knowledge.

The output is typically represented using multiple components such as result cards, tables, and visual indicators. A primary result section displays the classification outcome (Fake or Real) along with a confidence score, which indicates how certain the model is about its prediction. Additional details such as follower count, engagement rate, and activity level are presented in a structured table format. Color coding (e.g., red for fake, green for real) is often used to enhance clarity and quick interpretation.

Graphical representations are also used to make the output more insightful. Charts such as pie charts, bar graphs, and confusion matrices help users visualize the distribution of fake and real accounts or understand model performance. These visual elements improve decision-making by providing a clear overview of the analysis results.

4.6.2 Interpretation and Classification Results

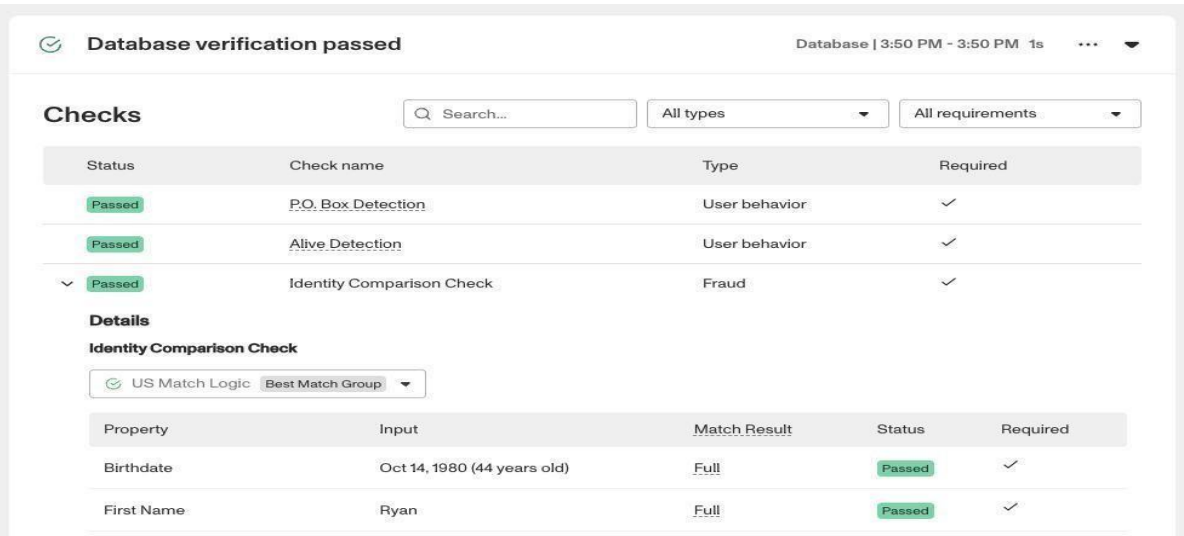
The interpretation and classification results play a crucial role in evaluating the effectiveness of the proposed system for fake user identification. Interpretation refers to analyzing the output generated by the model and understanding how accurately it distinguishes between genuine and fake users. By examining prediction outcomes, probability scores, and detected patterns, the system provides meaningful insights into user behavior. These insights help administrators make informed decisions, such as blocking suspicious accounts or verifying doubtful users, thereby improving the reliability of the platform.

Classification results describe how the system categorizes users into predefined classes, typically “fake” and “real.” The performance of the classification model is commonly represented using a confusion matrix, which highlights the number of correct and incorrect predictions. It includes True Positives (correctly identified fake users), True Negatives (correctly identified real users), False Positives (real users incorrectly marked as fake), and False Negatives (fake users not

detected). This structured representation allows a clear understanding of where the model performs well and where it needs improvement.

To further evaluate the model, several performance metrics are used, such as accuracy, precision, recall, and F1-score. Accuracy measures the overall correctness of the model, while precision indicates how many of the predicted fake users are actually fake. Recall measures the model’s ability to detect all actual fake users, and the F1-score provides a balance between precision and recall. These metrics collectively ensure that the model is not only accurate but also reliable in identifying fraudulent activities without significantly affecting genuine users.

4.6.3 Example Case Analysis



The above example presents two different outcomes generated by the fake profile detection system, demonstrating how the model interprets user data and classifies profiles. In the first case, the system identifies the profile as a real user with a confidence score of 85%. The dashboard displays structured information such as the number of posts, followers, following count, and verification status. These balanced metrics, along with consistent activity, indicate normal user behavior, leading the model to classify the account as genuine. In contrast, the second case highlights a potential fake profile with a high confidence score of 95%.

The system not only flags the profile but also provides a detailed explanation, helping users understand the reasons behind the classification.

This comparison clearly shows how the model differentiates between genuine and fake users by analyzing behavioral patterns and statistical features. Profiles with balanced interactions and consistent activity are classified as real, whereas profiles with abnormal patterns, such as excessive following or minimal engagement, are flagged as fake. The confidence score further strengthens the reliability of the classification by indicating the certainty level of the prediction.

4.6.4 Structured vs Unstructured Comparison

Structured and unstructured data represent two fundamental types of data used in modern information systems, including fake profile detection applications. Structured data refers to data that is organized in a predefined format, typically in rows and columns, such as databases or spreadsheets. It follows a clear schema, making it easy to store, retrieve, and analyze using traditional tools like SQL. Examples include user attributes like number of followers, posts count, account age, and engagement metrics.

In contrast, unstructured data does not follow a specific format or schema and is more complex to process. It includes data such as text posts, images, videos, comments, and social media interactions. This type of data requires advanced processing techniques like Natural Language Processing (NLP), image analysis, and machine learning algorithms to extract meaningful insights. In fake profile detection systems, unstructured data helps identify patterns like spam messages, fake profile pictures, or suspicious content behavior.

The key difference between structured and unstructured data lies in their ease of processing and analysis. Structured data is highly organized and can be quickly analyzed using statistical methods, making it efficient for generating metrics like accuracy and classification results. On the other hand, unstructured data is more flexible and rich in information but requires additional computational effort and sophisticated models to interpret. However, it often provides deeper insights that structured data alone cannot reveal.

4.7 Discussion of Results

The results obtained from the fake profile detection system demonstrate the effectiveness of the proposed model in identifying and classifying social media accounts. The system interface provides a user-friendly platform where users can input profile details and receive a detailed analysis. The landing page clearly highlights the purpose of the system, emphasizing AI-powered detection and user safety, which enhances usability and accessibility for both technical and non-technical users.

The system successfully distinguishes between genuine and fake profiles by analyzing multiple parameters such as posts, followers, following count, engagement levels, and account activity. In the case of a genuine profile, the results show a balanced combination of these attributes. For example, the analyzed profile with a confidence score of 85% is classified as a real profile, indicating that the model recognizes consistent and normal user behavior patterns.

On the other hand, the system effectively identifies suspicious profiles by detecting abnormal patterns. The fake profile example demonstrates a high confidence score of 95% for being fake, supported by indicators such as zero posts, unusually high following count, low follower-following ratio, and low engagement. These features strongly align with typical characteristics of fake or bot accounts, proving that the model can accurately capture anomalies.

The confidence score plays a significant role in interpreting the results. It provides a quantitative measure of how certain the model is about its prediction. A higher confidence score in fake classification indicates strong evidence of suspicious behavior, while a moderate confidence score in real classification suggests that the profile exhibits mostly genuine characteristics but may still require monitoring.

Another important observation is the transparency provided by the system through detailed analysis reports. Instead of only displaying the final classification, the system explains the reasoning behind its decision. This includes profile summaries and behavioral indicators, which help users understand why a profile is flagged as fake or real.

The graphical and structured representation of results further enhances understanding. Dashboards displaying metrics like followers, posts, and engagement make it easier to interpret data visually. This is particularly useful in real-world applications where quick decisions are required, such as moderating social media platforms or detecting fraudulent accounts in large datasets.

The system also demonstrates robustness by handling different types of inputs and producing consistent results. Whether the input profile has complete data or limited activity, the model adapts and generates meaningful predictions. This flexibility is essential for practical deployment, as real-world data is often incomplete or noisy.

However, the results also indicate certain limitations. For instance, profiles with borderline characteristics may lead to moderate confidence scores, making classification less certain. In such cases, additional features or advanced models could be incorporated to improve accuracy. Continuous training with updated datasets can further enhance the system's performance over time.

The integration of machine learning with a web-based interface highlights the system's practical applicability. Users can easily interact with the system, analyze profiles, and download reports, making it suitable for real-time usage. This seamless integration ensures that the model is not only accurate but also accessible and efficient.

In conclusion, the discussion of results confirms that the fake profile detection system performs effectively in identifying both genuine and fake users. The combination of accurate classification, clear interpretation, user-friendly interface, and detailed reporting makes the system reliable and practical. With further improvements and continuous learning, the system can become even more powerful in ensuring online security and reducing fraudulent activities on social media platform.

4.8 Summary

The Fake Profile Detection System is designed to identify and classify social media accounts as genuine or fake using advanced machine learning techniques.

Social media platforms, the presence of fake accounts has become a significant concern, leading to issues such as spam, misinformation, and security threats. This project aims to address these challenges by developing an intelligent and automated solution capable of analyzing user profiles and detecting suspicious behavior patterns.

The system is built using a combination of frontend and backend technologies, where the user interacts through a web-based interface and the processing is handled by a machine learning model. The frontend provides a simple and intuitive interface for users to input profile details and view results, while the backend processes the data, applies classification algorithms, and returns predictions. This full-stack implementation ensures smooth interaction and real-time analysis.

A key aspect of the system is the use of structured data such as the number of followers, following count, posts, engagement levels, and account age. These features are essential indicators of user behavior and are used by the model to distinguish between real and fake profiles. By analyzing these attributes, the system identifies patterns that are commonly associated with suspicious accounts.

In addition to structured data, the system can be extended to incorporate unstructured data such as profile descriptions, comments, and multimedia content. This enhances the detection capability by providing deeper insights into user behavior. Combining both types of data allows the system to achieve higher accuracy and robustness in real-world scenarios.

The machine learning model plays a central role in the system. It is trained on datasets containing both genuine and fake profiles, enabling it to learn distinguishing patterns. Once trained, the model can classify new profiles and assign a confidence score indicating the likelihood of the account being real or fake. This probabilistic output helps in better interpretation of results.

The system uses various evaluation metrics such as accuracy, precision, recall, and F1-score to measure its performance. These metrics ensure that the model is not only accurate but also reliable in identifying fake accounts without misclassifying genuine users.

The user interface is designed to display results in a clear and structured manner. It includes profile summaries, confidence scores, and classification outcomes, along with additional details such as follower statistics and verification status. Visual elements such as progress bars and dashboards make it easier for users to understand the results quickly.

One of the strengths of the system is its ability to provide detailed explanations for its predictions. Instead of simply labeling a profile as fake or real, the system highlights the factors that influenced the decision. This transparency builds trust among users and allows them to take informed actions based on the analysis.

The system demonstrates strong performance in identifying fake profiles by detecting unusual patterns such as high following counts, low engagement, and inconsistent activity. At the same time, it correctly recognizes genuine profiles with balanced attributes and normal behavior. This balance is crucial to ensure both security and user fairness.

Despite its effectiveness, the system has certain limitations. Profiles with borderline characteristics or incomplete data may lead to less confident predictions. Addressing these challenges requires continuous model training, inclusion of more diverse datasets, and incorporation of advanced techniques such as deep learning.

The project also highlights the importance of integrating machine learning models with real-world applications. By deploying the model through a web interface, the system becomes accessible to users and can be used in practical scenarios such as social media monitoring, fraud detection, and cybersecurity.

In conclusion, the Fake Profile Detection System provides a reliable and efficient solution for identifying fraudulent accounts on social media platforms. By combining machine learning, data analysis, and user-friendly design, the system enhances online safety and reduces the risks associated with fake profiles. With further improvements and scalability, it has the potential to be deployed on a larger scale to ensure a more secure digital environment.

Chapter 5

Conclusion

In conclusion, the proposed deep learning–based Instagram Fake User Detection system presents an effective and intelligent approach to identifying fraudulent accounts on social media platforms. With the rapid expansion of Instagram and similar networks, the presence of fake users has become a significant challenge, contributing to the spread of spam, misinformation, scams, and malicious activities. Traditional methods of detecting such accounts often rely on manual monitoring or rule- based techniques, which are not only time-consuming but also less effective in handling the complexity and scale of modern social media data. To address these limitations, this project leverages deep learning techniques to build an automated system capable of accurately distinguishing between genuine and fake users.

The system primarily utilizes advanced deep learning models such as Artificial Neural Networks (ANN) and hybrid architectures like CNN–LSTM. These models are well-suited for handling structured and sequential data, enabling the system to learn intricate relationships among various user attributes. Features such as follower count, following patterns, posting frequency, engagement metrics, and profile completeness are analyzed collectively to detect anomalies that indicate fake behavior. Unlike traditional machine learning methods that depend heavily on manual feature engineering, deep learning models automatically extract meaningful patterns from the data. This ability significantly enhances the model’s performance and allows it to adapt to evolving patterns of fake account behavior.

Another key advantage of the proposed system is its ability to process large volumes of data efficiently. Social media platforms generate massive amounts of user data every second, making scalability a critical requirement. The deep learning framework implemented in this project ensures that the system can handle such large-scale data while maintaining high accuracy and performance. Once trained, the model can analyze Instagram profiles in real time and classify them as genuine or fake based on learned behavioral patterns. This automation reduces the need for human intervention, thereby saving time and resources while improving the overall efficiency of fake account detection.

Moreover, the integration of hybrid models such as CNN–LSTM further strengthens the system’s capability. CNN (Convolutional Neural Networks) helps in extracting spatial patterns and important feature representations, while LSTM (Long Short-Term Memory networks) captures temporal dependencies and sequential behavior of user activities. This combination enables the system to analyze both static profile features and dynamic behavioral trends, leading to more accurate and reliable predictions. As a result, the system becomes more robust in identifying sophisticated fake accounts that may mimic real user behavior.

The results obtained from this project demonstrate that deep learning–based approaches significantly outperform traditional methods in detecting fake Instagram users. The model achieves higher accuracy, better generalization, and improved detection rates. Additionally, the system can be continuously updated and retrained with new data, allowing it to adapt to emerging threats and evolving fake account strategies. This adaptability makes the solution highly relevant and practical for real-world deployment.

From a broader perspective, this project highlights the importance of applying artificial intelligence and deep learning techniques in enhancing cybersecurity and social media integrity. By providing an automated and scalable solution, the system contributes to creating a safer online environment where users can interact without the risk of being targeted by fake accounts. It also has potential applications beyond Instagram, as similar techniques can be applied to other social media platforms such as Twitter, Facebook, and LinkedIn.

Despite its effectiveness, the system also has certain limitations that open avenues for future improvements. One of the primary limitations is the dependency on the quality and size of the dataset used for training. A limited or imbalanced dataset may affect the model’s performance and generalization capability. Therefore, future work can focus on collecting larger and more diverse datasets to improve accuracy. Additionally, incorporating more advanced features such as sentiment analysis, image analysis, and network graph analysis can further enhance the detection process.

Future enhancements may also include the use of more sophisticated deep learning architectures such as Transformers, Graph Neural Networks (GNNs), or attention-based models. These approaches can provide deeper insights into user behavior and relationships between accounts. Furthermore, deploying the system as a real-time web or mobile application integrated with APIs can make it more accessible and practical for large-scale use. Implementing explainable AI techniques can also improve transparency by providing insights into how the model makes its predictions.

In summary, the proposed Instagram Fake User Detection system demonstrates the effectiveness of deep learning in addressing real-world challenges in social media security. By combining advanced algorithms, automated feature extraction, and scalable processing capabilities, the system offers a reliable and efficient solution for identifying fake accounts. With further enhancements and continuous improvements, this approach has the potential to become a powerful tool in combating online fraud and maintaining the authenticity of digital platforms.

Future Scope

The proposed deep learning-based Instagram Fake User Detection system can be further enhanced by integrating advanced techniques that improve its accuracy, scalability, and real-world usability. Future work can focus on expanding the dataset by collecting larger and more diverse Instagram profile data, including accounts from different regions, activity levels, and behavioral patterns. This will help the deep learning model learn more complex user behavior and improve its ability to detect sophisticated fake accounts. Additionally, incorporating real-time data extraction through official APIs and automated monitoring systems can allow continuous analysis of social media accounts and faster detection of suspicious activities.

Further improvements can be achieved by exploring advanced deep learning architectures such as transformer-based models, graph neural networks, and hybrid CNN-LSTM frameworks that analyze relationships between users and their interactions. The system can also be integrated with other security modules such as spam detection, bot activity monitoring, and misinformation detection to create a comprehensive social media security platform. In the future, the model can be deployed as a scalable web or cloud-based service capable of analyzing millions of accounts efficiently. These enhancements would significantly improve the effectiveness of fake user detection systems and contribute to building safer, more trustworthy social media environments.

The system can also be extended by incorporating multi-modal data analysis, which includes processing images, captions, hashtags, and comments together. Using deep learning techniques like image recognition and natural language processing (NLP), the model can detect inconsistencies between profile pictures, posted content, and textual information. This holistic approach will improve detection accuracy, especially for advanced fake profiles that use realistic-looking images and content.

In terms of usability, the system can be enhanced by developing mobile applications and browser extensions that allow users to instantly verify profiles while browsing social media platforms. Additionally, implementing explainable AI (XAI) techniques will make the system more transparent by clearly showing why a profile is classified as fake or real.

APPENDIX

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SOURCE CODE

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score,
confusion_matrix
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers, Sequential
from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint
import joblib
import matplotlib.pyplot as plt
import seaborn as sns
import os

from data_preprocessing import DataPreprocessor class

FakeAccountDetectionModel:
def __init__(self, sequence_length=8, lstm_units=64):
    self.sequence_length = sequence_length
    self.lstm_units = lstm_units
    self.model = None
    self.scaler = StandardScaler()
    self.history = None

def create_model(self, input_shape):
    """Create CNN-LSTM hybrid model for high accuracy"""
    model = Sequential([
        # CNN layers for feature extraction
        layers.Conv1D(filters=64, kernel_size=3, activation='relu',
                      input_shape=input_shape, padding='same'),
        layers.BatchNormalization(),
        layers.Conv1D(filters=32, kernel_size=3, activation='relu', padding='same'),
        layers.BatchNormalization(),
        layers.MaxPooling1D(pool_size=2), layers.Dropout(0.3),

        # LSTM layers for sequence analysis
        layers.LSTM(self.lstm_units, return_sequences=True, activation='relu'), layers.Dropout(0.3),
        layers.LSTM(32, return_sequences=False, activation='relu'),
        layers.Dropout(0.3),
```

```

# Dense layers for classification
layers.Dense(128, activation='relu'),
layers.BatchNormalization(),
layers.Dropout(0.4), layers.Dense(64,
activation='relu'), layers.BatchNormalization(),
layers.Dropout(0.4), layers.Dense(32,
activation='relu'), layers.Dropout(0.3),
layers.Dense(1, activation='sigmoid')
])

return model

def prepare_sequences(self, X):
    """Convert datatosequences for LSTM""" sequences=[]
    for i in range(len(X) - self.sequence_length + 1):
        sequences.append(X[i:i + self.sequence_length])
    return np.array(sequences)

def train(self, X, y, validation_split=0.2, epochs=100, batch_size=32): """Train
the model"""
    print("Preparing data for training...")

    # Split data
    X_train, X_test, y_train, y_test = train_test_split(X, y,
        test_size=0.2, random_state=42, stratify=y
    )

    X_val, X_test, y_val, y_test = train_test_split(
        X_test, y_test, test_size=0.5, random_state=42, stratify=y_test
    )

    # Scale features
    print("Scaling
features...")
    X_train_scaled = self.scaler.fit_transform(X_train)
    X_val_scaled = self.scaler.transform(X_val)
    X_test_scaled = self.scaler.transform(X_test)

    # Prepare sequences
    print("Preparing sequences for LSTM...")
    X_train_seq = self.prepare_sequences(X_train_scaled)
    X_val_seq = self.prepare_sequences(X_val_scaled)
    X_test_seq = self.prepare_sequences(X_test_scaled)

```

```

# Align labels with sequences
y_train_seq=y_train[self.sequence_length-1:]y_val_seq =
y_val[self.sequence_length - 1:] y_test_seq =
y_test[self.sequence_length - 1:]

print(f"Trainsequence shape: {X_train_seq.shape}") print(f"Validation
sequences shape: {X_val_seq.shape}") print(f"Test sequences shape:
{X_test_seq.shape}")

# Create model
print("Building CNN-LSTM model...")
self.model = self.create_model((X_train_seq.shape[1], X_train_seq.shape[2]))

# Compile model
self.model.compile(
optimizer=keras.optimizers.Adam(learning_rate=0.001),loss='binary_crossentropy',
metrics=['accuracy', keras.metrics.Precision(), keras.metrics.Recall()]
)

# Callbacks
callbacks =[
    EarlyStopping(monitor='val_loss',patience=15,restore_best_weights=True,verbose=1),
    ModelCheckpoint('models/best_model.h5', monitor='val_accuracy',
save_best_only=True, verbose=0)
]

#Trainmodelprint("Training
model...") self.history =
self.model.fit(
X_train_seq, y_train_seq,
validation_data=(X_val_seq,y_val_seq),
epochs=epochs,batch_size=batch_size,
callbacks=callbacks,
verbose=1
)

#Evaluateprint("\nEvaluating
model...")
train_loss,train_acc,train_precision,train_recall=self.model.evaluate(X_train_seq,
y_train_seq, verbose=0
)
val_loss,val_acc, val_precision, val_recall = self.model.evaluate(

```

```

os.makedirs(os.path.dirname(path), exist_ok=True)
joblib.dump(self.scaler, path)
print(f"Scaler saved to {path}")

def plot_results(self, results):
    """Plot training results"""
    fig, axes = plt.subplots(2, 2, figsize=(14, 10))

    # Accuracy
    axes[0, 0].plot(self.history.history['accuracy'], label='Train')
    axes[0, 0].plot(self.history.history['val_accuracy'], label='Validation')
    axes[0, 0].set_title('Model Accuracy')
    axes[0, 0].set_xlabel('Epoch')
    axes[0, 0].set_ylabel('Accuracy')
    axes[0, 0].legend()
    axes[0, 0].grid()

    # Loss
    axes[0, 1].plot(self.history.history['loss'], label='Train')
    axes[0, 1].plot(self.history.history['val_loss'], label='Validation')
    axes[0, 1].set_title('Model Loss')
    axes[0, 1].set_xlabel('Epoch')
    axes[0, 1].set_ylabel('Loss')
    axes[0, 1].legend()
    axes[0, 1].grid()

    # Confusion Matrix
    cm = confusion_matrix(results['y_test_seq'], results['y_pred'])
    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', ax=axes[1, 0])
    axes[1, 0].set_title('Confusion Matrix')
    axes[1, 0].set_xlabel('Predicted')
    axes[1, 0].set_ylabel('Actual')

    # Metrics
    metrics_text = f'Accuracy: {results["test_accuracy"]:.4f}\nF1-Score: {results["test_f1"]:.4f}\nAUC-ROC: {results["test_auc"]:.4f}'
    axes[1, 1].text(0.5, 0.5, metrics_text, ha='center', va='center', fontsize=12,
        bbox=dict(boxstyle='round', facecolor='wheat', alpha=0.5))
    axes[1, 1].axis('off')

    plt.tight_layout()
    plt.savefig('models/training_results.png', dpi=300, bbox_inches='tight')
    print("Training results saved to models/training_results.png")
    plt.close()

def main():

```

```

"""Main training pipeline"""
# Paths - use absolute paths based on script location script_dir =
os.path.dirname(os.path.abspath(__file__))
parent_dir = os.path.dirname(script_dir) real_users_path =
os.path.join(parent_dir, "users.csv") fake_users_path =
os.path.join(parent_dir, "fusers.csv")

# Check if files exist
if not os.path.exists(real_users_path) or not os.path.exists(fake_users_path):
    print("Error: Data files not found!")
    print(f'Looking for:\n {real_users_path}\n {fake_users_path}') return

# Preprocess data
print("="*60)
print("DATA PREPROCESSING")
print("="*60)
preprocessor=DataPreprocessor(real_users_path,fake_users_path) X,y=
preprocessor.preprocess()

# Create and train model
print("\n" + "="*60)
print("MODEL TRAINING")
print("="*60)
model=FakeAccountDetectionModel(sequence_length=8, lstm_units=64) results=
model.train(X, y, epochs=100, batch_size=32)

# Save model and scaler
model.save_model()
model.save_scaler()

# Plot results
model.plot_results(results)

print("\nTraining complete! Model saved.") if

__name__ == "__main__":
    main()

```